

IMO(42nd) – 2001

First Day

1. Let ABC be an acute angled triangle with circum centre O. Let P on BC be the foot of the altitude from A. Suppose that $\angle BCA \geq \angle ABC + 30^\circ$. Prove that $\angle CAB + \angle COP < 90^\circ$.
2. Prove that $\frac{a}{\sqrt{a^2 + 8bc}} + \frac{b}{\sqrt{b^2 + 8ca}} + \frac{c}{\sqrt{c^2 + 8ab}} \geq 1$ for all positive real numbers a, b and c.
3. Twenty-one girls and twenty-one boys took part in a mathematical contest.
 - Each contestant solved at least six problems.
 - For each girl and each boy, at least one problem was solved by both of them.
 Prove that there was a problem that was solved by at least three girls and at least three boys.

Second Day

4. Let n be an odd integer greater than 1, and let $k_1, k_2, \dots, k_n$ be given integers. For each of the $n!$ permutations $a = (a_1, a_2, \dots, a_n)$ of $1, 2, \dots, n$, let $S(a) = \sum_{i=1}^n k_i a_i$. Prove that there are two permutations b and c , $b \neq c$ such that $n!$ is a divisor of $S(b) - S(c)$.
5. In a triangle ABC, let AP bisect $\angle BAC$, with P on BC, and let BQ bisect $\angle ABC$, with Q on CA. It is known that $\angle BAC = 60^\circ$ and that $AB + BP = AQ + QB$. What are the possible angles of triangle ABC?
6. Let a, b, c, d be integers with $a > b > c > d$. Suppose that $ac + bd = (b + d + a - c)(b + d - 1 + c)$ prove that $ab + cd$ is not prime.

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